MILITARY SPECIFICATION

INDICATOR, THRUST, SYNCHRO STYLE, PRESSURE RATIO SENSITIVE, 1.1-3.4 UNITS, 2 INCH SIZE, TYPE MR-2

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of synchro style pressure ratio thrust indicator designated Type MR-2.

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-I-7057	Indicator, Synchro, Aircraft, General Specification for
MIL-L-25142	Luminescent Material, Fluorescent
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument
MS33639	Case, Instrument, Clamp Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, handbooks, drawings, publications, and other government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19013.)

(Nongovernment standards and other publications are normally available from the organization which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specificaton, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3)

3.2 General specification. The requirements of MIL-I-7057 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.2.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

3.3 Design and construction. The indicator shall be desgined to indicate, at a remote station, the pressure ratio of a turbo-jet engine when used in conjunction with a pressure ratio transmitter wired in accordance with figure 2.

3.3.1 Case. The case shall conform to MS33639 and shall be hermetically sealed. The length of the case, excluding the connector, shall be 4.50 inches maximum.

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3.3.1.1 Filling tube. The filling tube, if used, shall be at least 1.25 inches long and shall not be visible when installed in the aircraft.

3.3.2 Dual synchros. The indicator shall contain two self-synchronous synchros geared or otherwise connected to two separate pointers. There shall be no mechanical connection between the output shafts of the two synchros.

3.3.3 Dial. The dial shall be fabricated of metal and shall be securely attached to the frame of the indicator mechanism. The dial shall conform to figure 2.

3.3.3.1 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.200 inch.

3.3.4 Pointer.

3.3.4.1 Main dial pointer. The pointer shall conform to MS33585-8, except that the taper at the point shall extend .3125 inch and the length of the pointer shall be such that the tip will overlap from 1/3 to 2/3 the length of the shortest graduation.

3.3.4.2 Sub dial pointer. The pointer shall conform to figure 2.

3.3.5 Markings.

3.3.5.1 Dial and pointer finish. The following markings shall be finished in fluorescent-luminescent material conforming to MIL-I-25142. The background of the dial and unshaded portion of the pointer shall be lusterless black, color number 37038 of FED-STD-595. The dimension of the markings shall be as listed below.

	Heights or Length Inch + .016	Width of Line or Graduation Inch + .005
Main Dial Numbers	.125	.020
Sub-Dial Numerals	.094	.016
Main Dial Graduation	.094	.016
Sub Dial Graduation	.063	.010
Lettering	.125	.020
Electrical Zero Reference	.031	.010
Shaded Portion of Pointer	-	-

3.3.6 Electrical receptacle. The electrical receptacle shall conform to MS33678R20-7P.

3.3.7 Wiring. Indicator internal and external wiring from indicator to transmitter shall be in accordance with figure 1.

3.3.8 Synthetic jewel bearings. When sapphire or ruby synthetic jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.4 Weight. The weight of the indicator shall not exceed 1.0 pound.

3.5 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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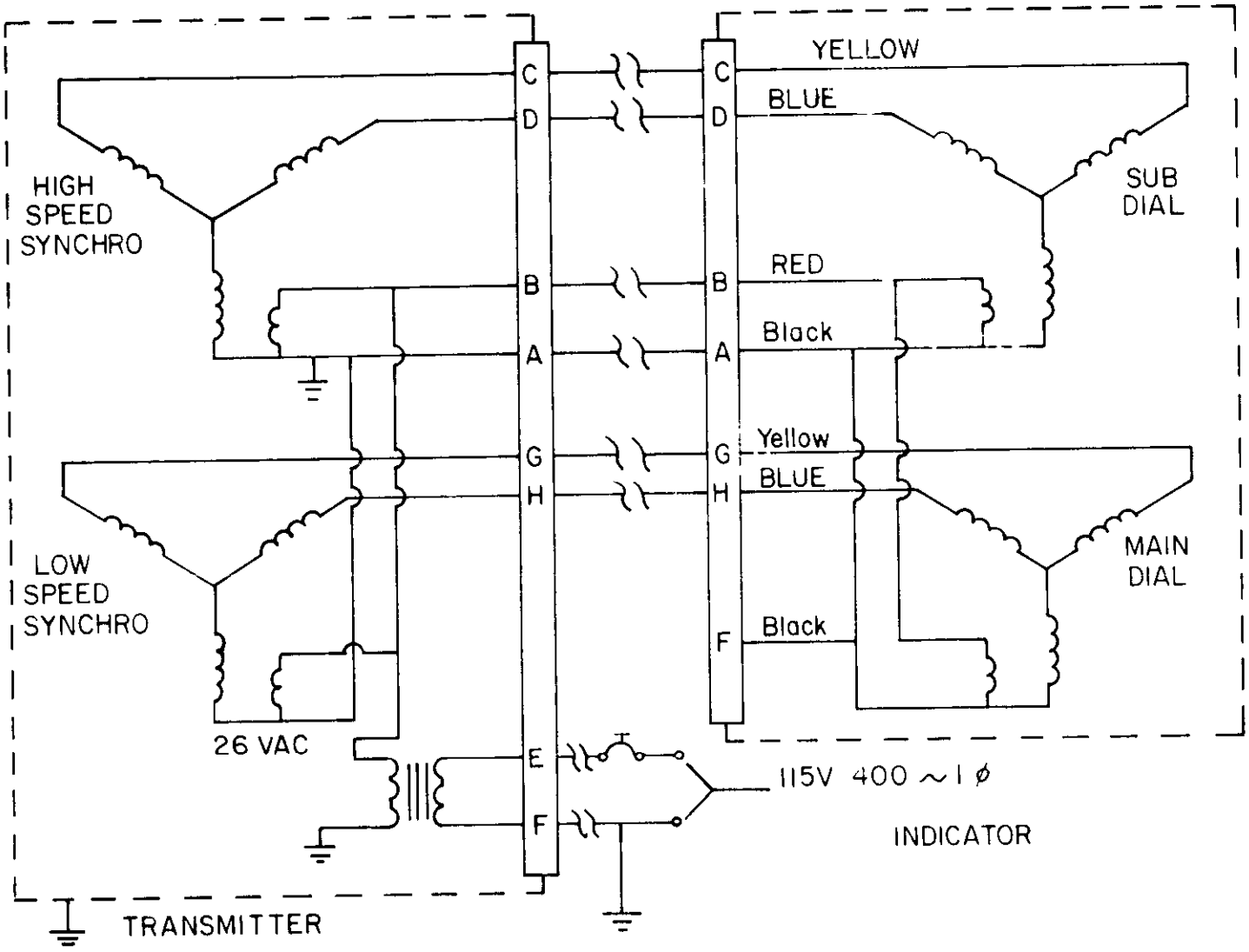


FIGURE 1 - Wiring diagram

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection sytem or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 General specification. The indicator shall be inspected and tested in accordance with MIL-I-7057 and as specified herein.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4)
b. Quality Conformance inspection (see 4.5)

4.4 Qualification inspection.

4.4.1 Test samples. The qualification test samples shall consist of three indicators representative of production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831.

4.4.3 Qualification tests. The qualification tests shall consist of all the tests of this specification including the test of MIL-I-7057.

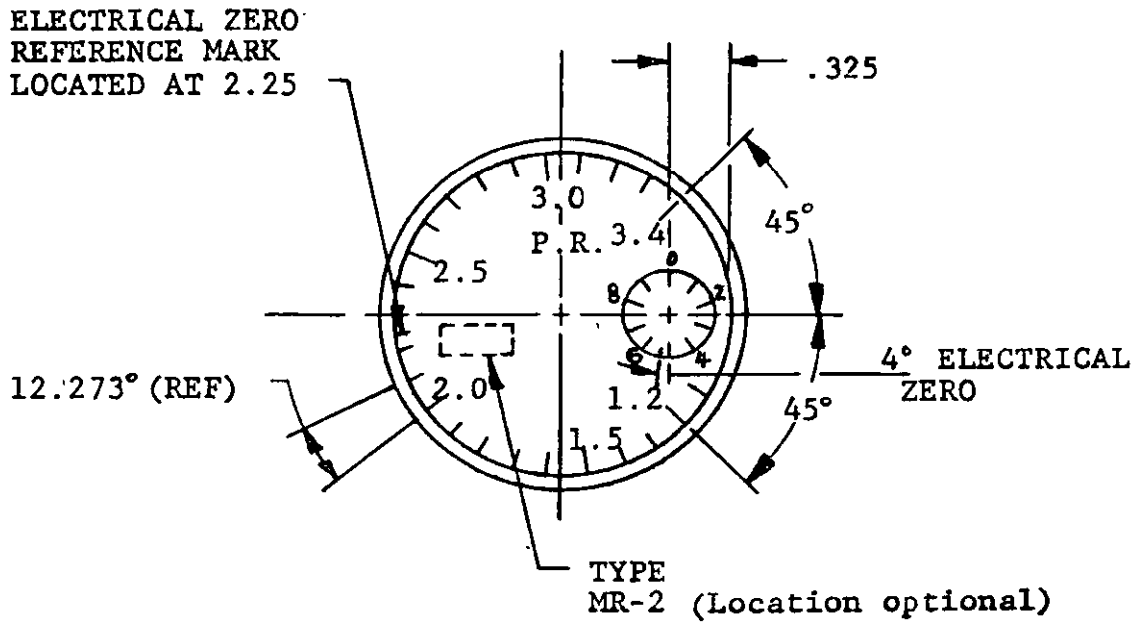
4.5 Quality conformance inspection. Quality conformance inspection shall consist of individual inspection and sampling inspection. Tests shall be in accordance with MIL-I-7057 and as specified herein. The test described under 4.6.7 shall replace the fogging test of MIL-I-7057.

4.6 Test methods.

4.6.1 Electrical zero. Connections between the indicator and the electrical zero transformer shall conform to figure 3.

4.6.2 Scale error and friction. The indicator shall be tested for scale error and

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DIAL LAYOUT

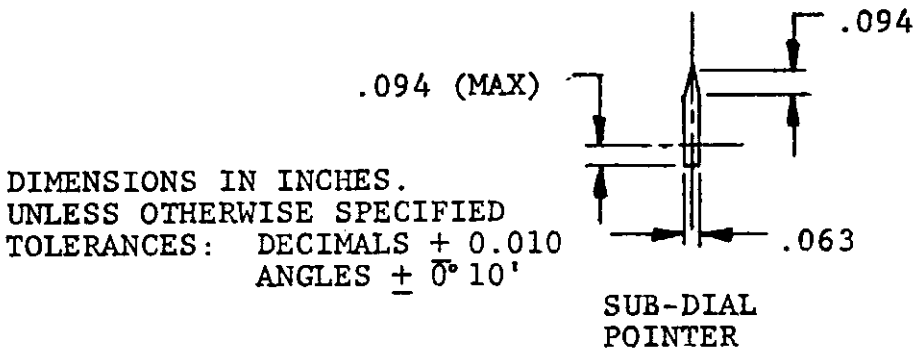


FIGURE 2. Details, dial and pointer.

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friction at the test points specified in Table I. Scale error and friction error shall not exceed the tolerances specified in Table II.

TABLE I

TEST POINTS

Main Dial Low Speed		Sub-Dial High Speed	
Ratio	Degree	Ratio	Degrees
1.05	32.591		
1.10	38.727		
1.20	51.000	0.01	32.00
1.50	87.818	0.02	68.00
1.80	124.636	0.03	104.00
2.00	149.182	0.04	104.00
2.20	173.727	0.05	176.00
2.30	186.000	0.06	212.00
2.40	198.273	0.07	248.00
2.60	222.818	0.08	284.00
3.00	271.909	0.09	320.00
3.40	321.000	0.00	356.00

TABLE II

TOLERANCES

Test	Pressure Main Dial	Ratio Units Sub-Dial
Scale error		
Room temp	+ 0.020	+ 0.001
High temp	+ 0.020	+ 0.001
Low temp	+ 0.020	+ 0.001
Position error	+ 0.006	+ 0.001
Friction error		
Room temp	+ 0.010	+ 0.001
High temp	+ 0.010	+ 0.001
Low temp	+ 0.016	+ 0.001
Vibration		
Pointer oscillation	+ 0.006	+ 0.001
Pointer variation	+ 0.006	+ 0.001

4.6.3 Position error. The tolerance specified in Table II shall apply.

4.6.4 Vibration error. Pointer oscillation and variation shall not exceed the tolerances specified in Table II.

4.6.5 Low temperature. Scale and friction errors shall not exceed the tolerances specified in Table II. The indicator shall not require more than 5 seconds to come within the specified tolerance.

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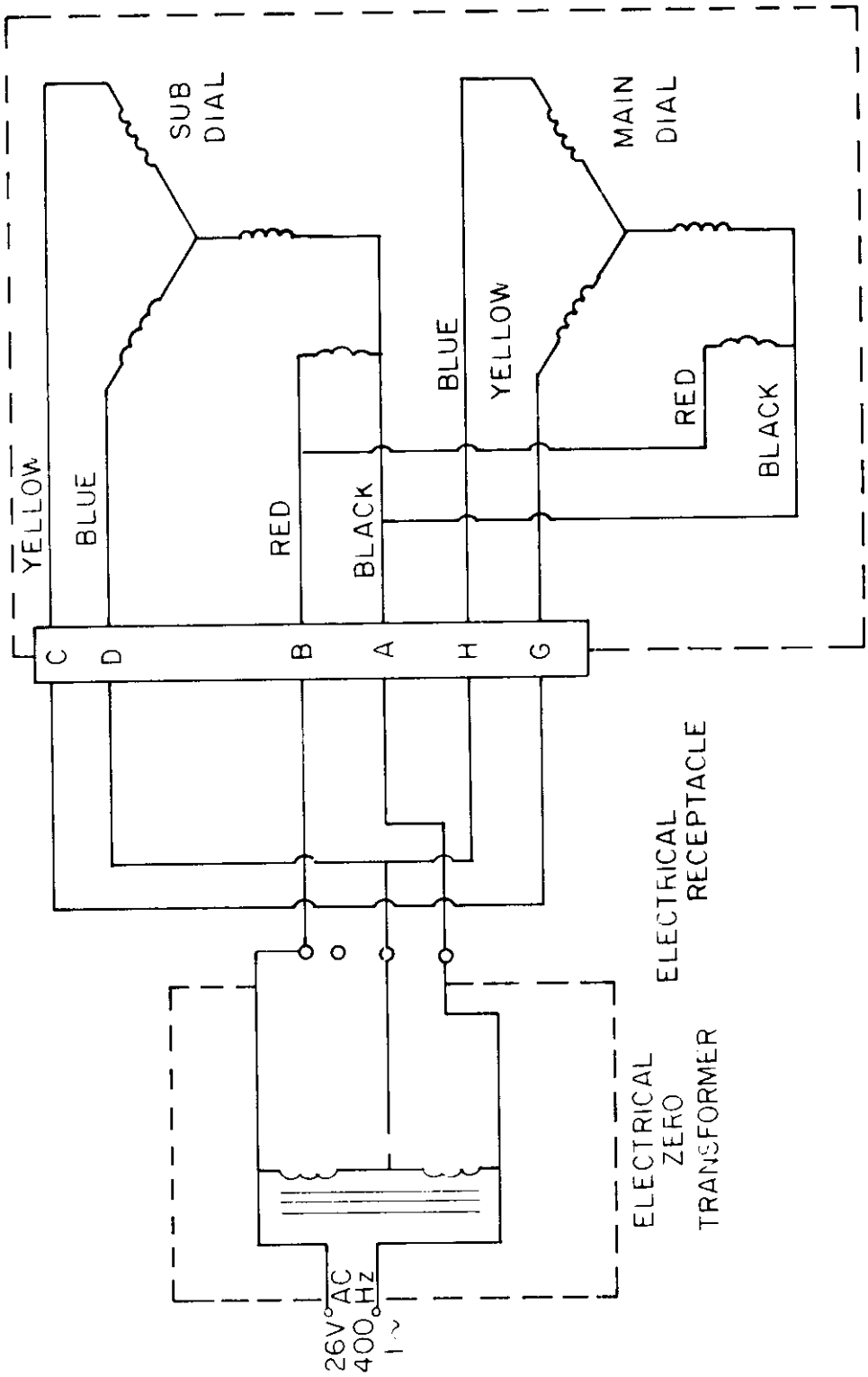


FIGURE 3 Electrical Zero Test Connections

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4.6.6 High temperature. Scale and friction errors shall not exceed the tolerances specified in Table II.

4.6.7 Fogging. The indicator shall be placed in a $90^{\circ} + 5^{\circ}C$ controlled ambient temperature for a minimum of 2 hours. The indicator shall then be removed from the high temperature chamber and within 30 seconds the indicator face shall be immersed to a depth of 1 to 2 inches in a water bath maintained at a temperature of 0° to $5^{\circ}C$. After 3 minutes the indicator shall be removed from the cold water and dried with air from a compressed air line. Any evidence of oil water fog inside the cover glass shall be cause for rejection.

4.6.8 High altitude-low temperature. The high altitude-low temperture test shall not be conducted.

4.6.9 Low temperature exposure. The low temperture and room temperature scale and friction error tolerances of Table II shall apply.

4.6.10 High temperature exposure. The high temperature and room temperature scale and friction error tolerances of Table II shall apply.

4.7 Inspection of preparation for delivery. Inspection of the preservation, packaging, and packing shall be conducted in accordance with the quality assurance provisions of MIL-STD-794 to insure, compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation-application. Preservation-packaging shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicators shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method 1C1 of MIL-P-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the tube when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of indicator shall be in accordance with ASTM D3951.

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5.2 Packing. Packing shall be level A, B, C, or Indsutrial as specified (see 6.2).

5.2.1 Level A. Indicators shall as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source tothe first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicators shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type MR-2 indicator covered by this specification is intended for use to indicate aircraft turbo-jet engine pressure ratio thrust measurements.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. When sampling plan B tests will not be conducted (see MIL-I-7057)
c. Level of packaging and packing required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the application Qualified Products List whether or not such products have actually been so listed by the date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center, MMEDO, Tinker AFB, Oklahoma 73145-5990, and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word) listing.
Aircraft jet engine
Dual synchros
Fluorescent-luminescent markings

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Hermetically sealed
Indicator
Length 4.50 inches maximum
Pressure ratio thrust

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with repsect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 71

Project No:
6620-F408

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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